

Last Time

Dispersion

Least Squares Principle

Skewness and Kurtosis

Outliers

Z-scores

Probability Basics

- Vocabulary

- Axioms and Rules

- Determining Probabilities

- Joint and Conditional Events

Permutations and Combinations

Today

Probability Continued

- Axioms and Rules

- Determining Probabilities

- Joint and Conditional Events

Permutations and Combinations

Random Variables

Distributions

- Discrete Distributions

Probability

Consider tossing a coin. Terms to know when talking about probability include:

1. Event: what comes up on the toss, heads (H) or tails (T)
2. Experiment: the act of tossing the coin
3. Joint event: if the coin is tossed more than once, the sequence of observations $\{H, T, T, H, H, \dots\}$
4. Sample space: the set of possible outcomes $S = \{H, T\}$

Axioms

Let the sample space S for a variable X be $S = \{E_1, E_2, \dots, E_m\}$. Probability $Pr()$ is a *measure* assigned to events $E_i \in S$ such that these three axioms hold:

1. $0 \leq Pr(E_i) \leq 1$
2. $Pr(S) = 1$, and
3. For *mutually exclusive events*, $Pr(E_i \text{ or } E_j) = Pr(E_i) + Pr(E_j)$.

The events $\{E_1, E_2, \dots, E_m\}$ are called *elementary events*.

Set Operations

We use set operations to describe/perform different probability computations.

1. Union (or, $\cup$)
2. Intersection (and, $\cap$)
3. Complement (not, indicated as $\bar{A}$, $\sim A$, or A^C)

Probability Rules

From the axioms we can show that

1. $Pr(\bar{S}) = Pr(\emptyset) = 0$
2. $Pr(\bar{A}) = 1 - Pr(A)$
3. $Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$
4. $Pr(A) = Pr(A \cap B) + Pr(A \cap \bar{B})$ (Law of Total Probability)

How do we assign probabilities?

1. From a MODEL: a pretty good idea of what probabilities should be
2. By doing the experiment a large number of times: the frequency approach.

Bernoulli's Theorem

What is large?

Let N be the number of experiments, n_i be the number of times outcome E_i was observed, and p_i be the “true” probability $Pr(E_i)$.

$$\lim_{N \rightarrow \infty} Pr(|p_i - n_i/N| > \delta) = 0 \text{ for } \delta > 0$$

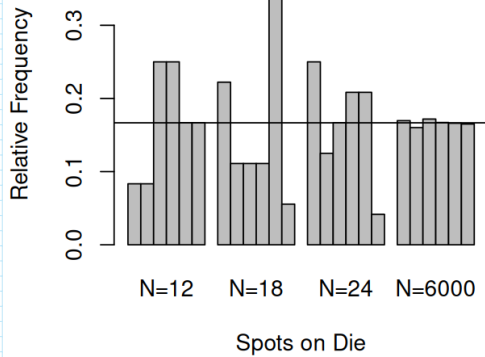
In R

```
N <- c(12,18,24,6000)
die <- NULL

for (i in 1:length(N))
  die <- rbind(die,hist(sample(6,N[i],replace=TRUE),
    breaks=seq(.5,6.5,by=1))$counts/N[i])

barplot(t(die),
  beside = TRUE, col = "gray",
  names.arg = c("N=12", "N=18", "N=24","N=6000"),
  ylab = "Relative Frequency", xlab = "Spots on Die")
abline(h=.166667,col="black")
```

Bernoulli's Theorem



Simple Die Example

Roll a fair die.

Elementary events: $S = \{1, 2, 3, 4, 5, 6\}$

Simple Die Example

Roll a fair die.

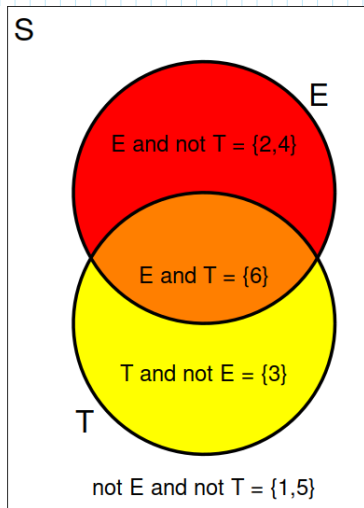
Elementary events: $S = \{1, 2, 3, 4, 5, 6\}$

Compound events:

- ▶ $O = \{1, 3, 5\}$ (an odd number)
- ▶ $E = \{2, 4, 6\}$ (an even number), and
- ▶ $T = \{3, 6\}$ (divisible by 3).

Compute $Pr(O)$, $Pr(E)$, $Pr(T)$ and $Pr(\overline{T})$

Joint Events



Simple Dice Example

Toss a red die and a green die. Let R = spots on red die, G = spots on green die

- ▶ What is the sample space?
- ▶ What are the elementary events in the sample space?
- ▶ What is the probability of each possible outcome?
- ▶ What is the probability that $R = G$?
- ▶ What is the probability that either $R = 6$ or $G = 6$?

Conditional Probabilities

GSS Respondents: Young (< 50) $\times$ Attitude

Young	Att	
	1=Yes	2=No
FALSE	122	102
TRUE	215	434
Total		873

Joint Probability Table

Remember: Joint events are defined by the co-occurrence of two single events.

	Att		
Young	1=Yes	2=No	
FALSE	.140	.117	.257
TRUE	.246	.497	.743
	.386	.614	

Conditional Probabilities

How does our knowledge of one event (e.g., a person is 50 years old or older: Young=FALSE) change our expectations about the outcome of another event (e.g., endorsing the attitude that young people should respect the opinions of their elders: Att=1)?

Marginal probability: $Pr(\text{Att} = 1) = .386$

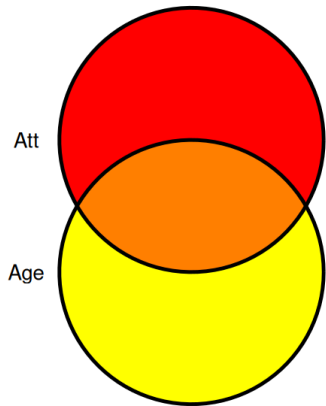
Conditional probability: $Pr(\text{Att} = 1 | \text{Young} = \text{FALSE})$

Independent Events

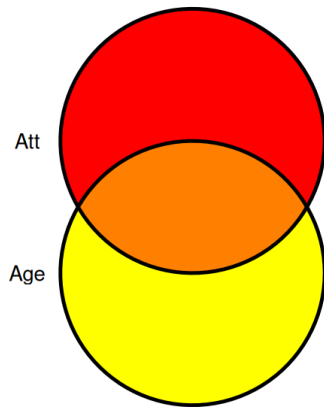
If events Young and Att are independent, knowing Young=FALSE will not change my expectations that Att=1:

$$Pr(\text{Att} = 1 | \text{Young} = \text{FALSE}) = Pr(\text{Att} = 1)$$

Definition



Definition



$$Pr(\text{Att} = 1 | \text{Young} = \text{FALSE}) = \frac{Pr(\text{Att} = 1 \cap \text{Young} = \text{FALSE})}{Pr(\text{Young} = \text{FALSE})}$$

Back to Independent Events

If

$$Pr(\text{Att} = 1 | \text{Young} = \text{FALSE}) = Pr(\text{Att} = 1)$$

then

$$\frac{Pr(\text{Att} = 1 \cap \text{Young} = \text{FALSE})}{Pr(\text{Young} = \text{FALSE})} = Pr(\text{Att} = 1)$$

Fundamental Counting Principle

If there are n possible outcomes for one experiment and m possible outcomes for another, then there are $n \times m$ possible joint outcomes for the two experiments.

Example: I own $n = 5$ shirts and $m = 3$ pants. Disregarding bad style/color choices, how many different outfits can I wear?

Example: Roll two dice of different colors. The number of unique outcomes is $6 \times 6 = 36$.

How to count up events

The way we determine how likely an event is is to

1. Write down the simplest sample space possible for our experiment and
2. Count up the elementary events in the sample space that correspond to the event of interest.

Sampling With vs. Without Replacement

- ▶ With replacement: An element of the sample space can appear in a sample more than once. Example: flipping a coin or rolling a die
- ▶ Without replacement: When an element is sampled, it is removed from the sample space. Example: dealing cards from a deck

When the sample space is infinite, sampling without replacement can be treated as sampling with replacement.

Permutations

Suppose there are 14 people in a wedding party that need to be seated at the “head” table at the reception. The newlyweds will sit in the two center seats with 6 attendants for Spouse 1 to the left and 6 attendants for Spouse 2 to the right.

1. How many ways can Spouse 1's attendants be seated to the left of Spouse 1?
2. How many ways can Spouse 2's attendants be seated to the right of Spouse 2?
3. How many ways in total can Spouse 1 and 2's attendants be seated?
4. Suppose the 12 attendants have equally strong attachments to both Spouse 1 and Spouse 2, so there is no reason to assign them to either side of the table. How many seating arrangements are there using both the left and right sides of the table?

Permutations, cont.

Suppose you're playing straight poker with a standard 52 card deck. How many different ways could your hand of 5 cards be dealt to you? This means that a deal of A,K,Q,J,10 is different from a deal of 10,A,K,Q,J, so order matters.

Combinations

You're still playing straight poker. How many different unique hands can you be dealt? If you have 4 of a kind in aces, it doesn't matter the order in which the aces and the odd card appear.

Binomial Coefficient

$$\binom{N}{x} = \frac{N!}{x!(N-x)!}$$

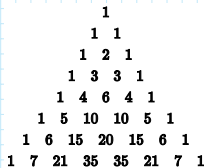
Binomial Coefficient

$$\binom{N}{x} = \frac{N!}{x!(N-x)!}$$

$$\sum_{x=0}^N \binom{N}{x} = 2^n$$

$$(a+b)^N = \binom{N}{0}a^N + \binom{N}{1}a^{N-1}b + \binom{N}{2}a^{N-2}b^2 + \cdots + \binom{N}{j}a^{N-j}b^j + \cdots + \binom{N}{N-1}ab^{N-1} + \binom{N}{N}b^N$$

Pascal's Triangle:



1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1

Some Exercises

Powerball lottery:

1. 5 numbers (1-69) are drawn without replacement from one machine
2. 1 (red) number (1-26) is drawn from the “Powerball” machine
3. Players must match all 5 numbers (in any order) and the red Powerball number

Compute:

1. How many elements are in the sample space?
2. What is the probability of matching the Powerball?
3. What is the probability of matching 1 of 5 and the Powerball, and not the Powerball?
4. What is the probability of matching n of the 5 ($n = 1, 2, \dots, 5$) and the Powerball, and not the Powerball?

Random Variable

A random variable X is a mapping between a sample space S and the real numbers $\mathcal{R}$:

$$X : S \rightarrow \mathcal{R}.$$

Discrete Random Variables

For a discrete random variable X :

- ▶ The sample space S is finite or countably infinite.
- ▶ Probability function
 1. $0 \leq p_X(r) \leq 1.0$ for all $r \in \mathcal{R}$
 2. $\sum_{r \in S} p_X(r) = 1.0$
 3. $Pr(X = r) = p_X(r)$

Continuous Random Variables

For a continuous random variable X :

- ▶ The *support* S is uncountably infinite (range(s) of real numbers).
- ▶ Probability density function
 1. $f_X(r) \geq 0$ for all $r \in \mathcal{R}$
 2. $\int_{-\infty}^{\infty} f_X(r) dr = 1.0$
 3. $Pr(X = r) = 0$

Cumulative Distributions

$$P_X(r) = \sum_{x \leq r} p_X(x)$$

$$F_X(r) = \int_{-\infty}^r f_X(x) dx$$

Expected Value

The population mean μ

Discrete:

$$E(X) = \sum_{x \in S} xp_X(x)$$

Continuous:

$$E(X) = \int_{-\infty}^{\infty} xf_X(x)$$

- ▶ $E(c) = c$
- ▶ $E(X + c) = E(X) + c$
- ▶ $E(cX) = cE(X)$

Expected Value

The population mean μ

Discrete:

$$E(X) = \sum_{x \in S} xp_X(x)$$

Continuous:

$$E(X) = \int_{-\infty}^{\infty} xf_X(x)$$

- ▶ $E(c) = c$
- ▶ $E(X + c) = E(X) + c$
- ▶ $E(cX) = cE(X)$

$$E(aX + bY) = aE(X) + bE(Y)$$

Variance

$$\text{Var}(X) = E[(X - \mu)^2] = E[(X - E(X))^2]$$

- ▶ $\text{Var}(c) = 0$
- ▶ $\text{Var}(X + c) = \text{Var}(X)$
- ▶ $\text{Var}(cX) = c^2 \text{Var}(X)$

Variance

$$\text{Var}(X) = E[(X - \mu)^2] = E[(X - E(X))^2]$$

- ▶ $\text{Var}(c) = 0$
- ▶ $\text{Var}(X + c) = \text{Var}(X)$
- ▶ $\text{Var}(cX) = c^2 \text{Var}(X)$

If X and Y are independent,

$$\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$$

Moments

	Raw	Central	Related Statistic
1	$E(X)$	$E(X)$	Mean
2	$E(X^2)$	$E[(X - E(X))^2]$	Variance
3	$E(X^3)$	$E[(X - E(X))^3]$	Skewness
4	$E(X^4)$	$E[(X - E(X))^4]$	Kurtosis

Distributions

Discrete	Continuous
Uniform	Uniform
Bernoulli	Normal
Binomial	Student's t
Poisson	χ^2
Tabled	F

R “Quads”

Distribution “root” name: norm

dnorm	probability (density) function
pnorm	cumulative distribution function
qnorm	quantile function
rnorm	random sample

R “Quads”

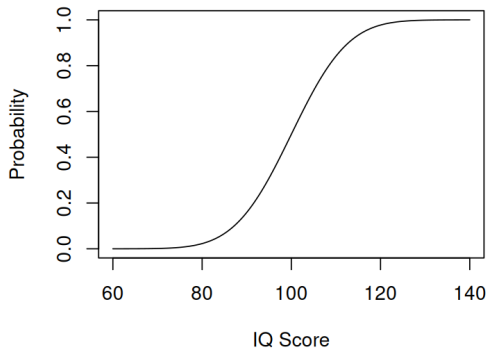
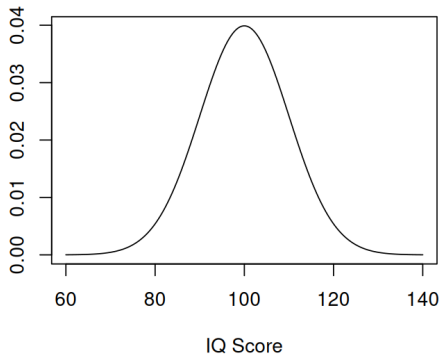
```
# Using R's distribution functions

# Choose a mean and standard deviation
# IQ scores have mean 100 and sd 10
mu <- 100
sigma <- 10

# Select IQ scores for plotting
iq <- seq(mu-4*sigma,mu+4*sigma,length=200)

# Plot the density and distribution functions
par(mfrow=c(1,2))
plot(iq,dnorm(iq,mu,sigma),
     ylab="Density",xlab="IQ Score",type='l')
plot(iq,pnorm(iq,mu,sigma),
     ylab="Probability",xlab="IQ Score",type='l')
par(mfrow=c(1,1)) #reset to single plots
```

R “Quads”

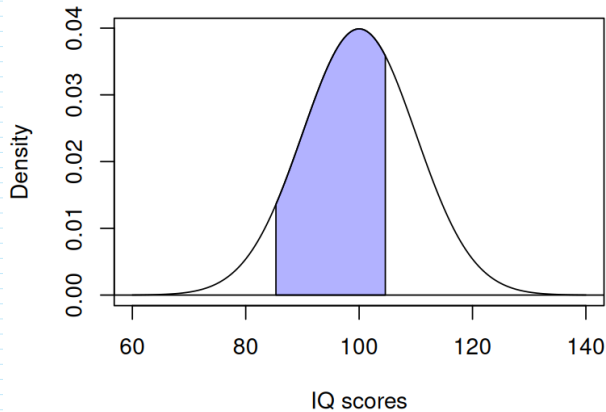


R “Quads”

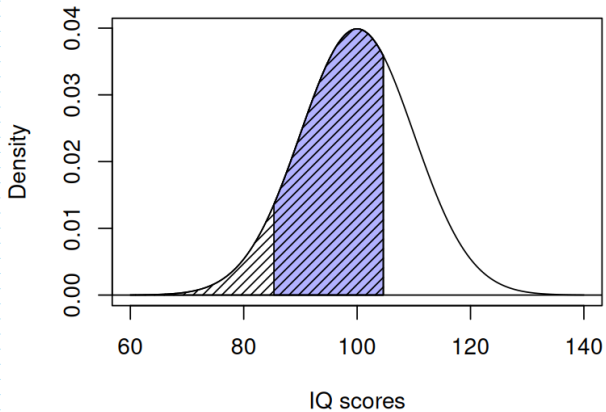
```
# Quantiles
p <- seq(0,1,by=.1)
cbind(100*p,qnorm(p,mu,sigma))

# Probabilities
# 1. Pr(IQ < 110)
pnorm(110,mu,sigma)
# 2. Pr(IQ > 95)
1 - pnorm(95,mu,sigma)
# 3. Pr(85 < IQ < 105)
pnorm(105,mu,sigma) - pnorm(85,mu,sigma)
```

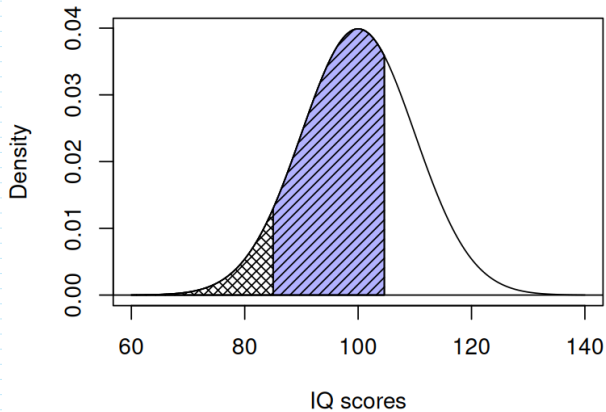
Probability as Area



Probability as Area



Probability as Area

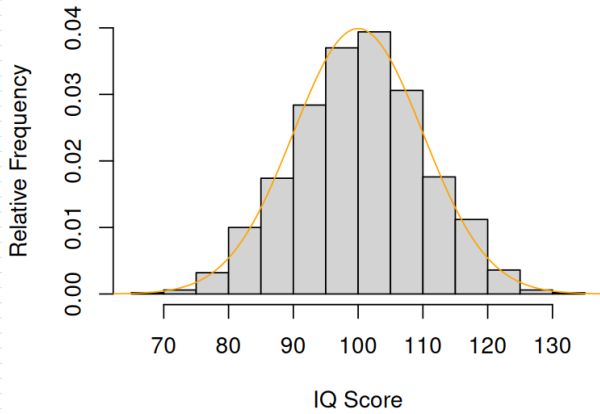


R “Quads”

```
# Random sampling
norm.sample <- rnorm(1000,mu,sigma)

hist(norm.sample,freq=FALSE,ylab="Relative Frequency",
      xlab="IQ Score",main="",ylim=c(0,.04))
lines(iq,dnorm(iq,mu,sigma),col='orange')
```

R “Quads”



Uniform

Roll a die.

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$m = |S| = 6$$

$$p_X(x|m) = \frac{1}{m} \text{ for } x \in S, 0 \text{ else}$$

Uniform

For consecutive integers from a to b ,

$$S = \{a, a+1, \dots, b-1, b\}$$

$$m = |S| = b - a + 1$$

$$p_X(x|m) = \frac{1}{b - a + 1} \text{ for } x \in S, \text{ 0 else}$$

Uniform

For consecutive integers from a to b ,

$$S = \{a, a+1, \dots, b-1, b\}$$

$$m = |S| = b - a + 1$$

$$p_X(x|m) = \frac{1}{b-a+1} \text{ for } x \in S, 0 \text{ else}$$

$$E(X|a, b) = \frac{a+b}{2}$$

$$\text{Var}(X|a, b) = \frac{(b-a+1)^2 - 1}{12}$$

Bernoulli

A single experiment with only two possible outcomes: success and failure.
Flip a coin, call “heads” a success.

$$\begin{aligned} S &= \{0, 1\} \\ 0 &\leq p \leq 1 \\ p_X(x|p) &= \begin{cases} p & x = 1 \\ 1 - p & x = 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Bernoulli

A single experiment with only two possible outcomes: success and failure.
Flip a coin, call “heads” a success.

$$\begin{aligned} S &= \{0, 1\} \\ 0 &\leq p \leq 1 \\ p_X(x|p) &= \begin{cases} p & x = 1 \\ 1 - p & x = 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

$$\begin{aligned} E(X) &= p \\ \text{Var}(X) &= p(1 - p) \end{aligned}$$

Binomial

Add up a sequence of N independent Bernoulli (0 or 1) experiments: flip a coin N times, answer N true/false questions. How many “successes” were there in the sequence?

$$S = \{0, 1, \dots, N\}$$

$$0 \leq p \leq 1$$

$$N \geq 1, N \in \mathcal{N}^+$$

$$p_X(x|N, p) = \begin{cases} \binom{N}{x} p^x (1-p)^{N-x} & x \in S \\ 0 & \text{otherwise.} \end{cases}$$

Binomial

Add up a sequence of N independent Bernoulli (0 or 1) experiments: flip a coin N times, answer N true/false questions. How many “successes” were there in the sequence?

$$S = \{0, 1, \dots, N\}$$

$$0 \leq p \leq 1$$

$$N \geq 1, N \in \mathcal{N}^+$$

$$p_X(x|N, p) = \begin{cases} \binom{N}{x} p^x (1-p)^{N-x} & x \in S \\ 0 & \text{otherwise.} \end{cases}$$

$$E(X) = Np$$

$$\text{Var}(X) = Np(1-p)$$

Poisson

- ▶ The number of rare events occurring in a fixed interval of time or bounded region of space. Airline accidents, hospital admissions, school shootings, typos on a web page, defective widgets in a shipment, etc.
- ▶ The limiting distribution of the binomial when $N \rightarrow \infty$ and $p \rightarrow 0$.

Poisson

- ▶ The number of rare events occurring in a fixed interval of time or bounded region of space. Airline accidents, hospital admissions, school shootings, typos on a web page, defective widgets in a shipment, etc.
- ▶ The limiting distribution of the binomial when $N \rightarrow \infty$ and $p \rightarrow 0$.

$$S = \{0, 1, 2, \dots\}$$

$$\lambda > 0$$

$$p_X(x|N, p) = \begin{cases} \frac{\lambda^x e^{-\lambda}}{x!} & x \in S \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \lambda$$

$$Var(X) = \lambda$$

Tabled

A random variable that can take on values in a list, with each value associated with a probability. Relative frequency distributions are examples of tabled probability distributions.

GSS Health Rating Responses

x	1	2	3	4
p_x	0.37	0.46	0.15	0.02

Tabled

A random variable that can take on values in a list, with each value associated with a probability. Relative frequency distributions are examples of tabled probability distributions.

GSS Health Rating Responses

x	1	2	3	4
p_x	0.37	0.46	0.15	0.02

$$S = \{1, 2, 3, 4\}$$

$$p_X(x|p_1, p_2, p_3, p_4) = \begin{cases} p_1 & x = 1 \\ \vdots & \\ p_4 & x = 4 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \sum_{x=1}^N x p_X(x)$$

$$Var(X) = \sum_{x=1}^N (x - E(X))^2 p_X(x)$$